

Assignment 3 — External SOAP Partner Integration

Multi-Agent AI Orchestration Platform

TeamID: Not provided in the supplied material

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1. Context

Partner edge: Enterprise SMS Gateway. The platform can notify a user when a long-running AI task finishes, fails, or requires attention. The external SMS gateway is represented as a SOAP partner, while the platform's own services remain REST-oriented. SOAP is justified at this boundary because the partner contract is strongly described by WSDL, the message structure is explicit, and the integration can apply message-level credentials and standardized SOAP faults. The one operation integrated here is **sendSMS**.

2. Partner WSDL

The supplied **partner.wsdl** defines all required WSDL elements: types, messages, portType, SOAP-over-HTTP binding, service, and port. The service endpoint is a realistic design-time placeholder:

https://sms.partner.example.com/soap/SmsGateway.

3. SOAP Request, Response and Fault

The request places partner credentials in the SOAP Header and the sendSMS call in the SOAP Body. The success response returns a provider message identifier and status. The fault models a realistic temporary gateway failure. The complete messages are submitted separately as XML files.

4. HTTP Binding

```
POST /soap/SmsGateway HTTP/1.1
Host: sms.partner.example.com
Content-Type: text/xml; charset=utf-8
SOAPAction: "https://partner.example.com/sms/sendSMS"

<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:sms="https://partner.example.com/sms">
  <soap:Header>...credentials...</soap:Header>
  <soap:Body>
    <sms:SendSMSRequest>...</sms:SendSMSRequest>
  </soap:Body>
</soap:Envelope>
```

5. Discovery — Modern Catalogue / Registry

At design time, the platform would obtain the partner WSDL either from a direct HTTPS WSDL URL supplied by the partner or from an internal integration catalogue. No live UDDI server is required. A catalogue record can point consumers to the service endpoint and the WSDL contract.

Field	Registry Entry
Business	Enterprise SMS Gateway
Service	EnterpriseSmsGateway
Endpoint	https://sms.partner.example.com/soap/SmsGateway
WSDL / tModel-style pointer	https://sms.partner.example.com/wsdl/SmsGateway.wsdl
Operation	sendSMS

6. Fault Mapping into Our Contract

The partner's vocabulary must not leak into the Multi-Agent platform's own contract. For example, the partner SOAP fault **GATEWAY_UNAVAILABLE** is mapped internally to the platform-level error **notification_unavailable**. The client of our platform therefore sees our own stable error vocabulary rather than the partner's provider-specific code.

Partner SOAP fault	Our platform contract error
GATEWAY_UNAVAILABLE	notification_unavailable
Authentication failure	notification_auth_failed
Invalid recipient/message	invalid_notification

Submission files

- integration.pdf
- partner.wsdl
- soap-request.xml
- soap-response.xml
- soap-fault.xml